

WELCOME NTI TO OUR DATA ANALYTICS PROJECT





NTI

Project Overview

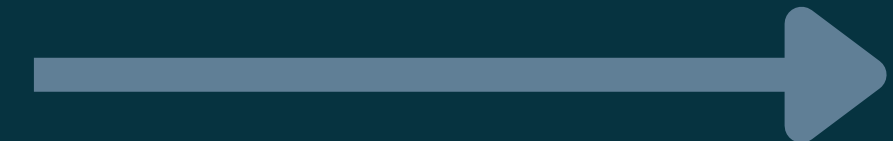
OUR DATASET IS ABOUT :

TELECOM COMPANY CUSTOMER CHURN PREDICTION

What's it's mainly about ?

The Customer Churn Prediction dataset is about a telecom company studying customer behavior ,demographics,services,contract type,tenure..etc . To measure if there's any business problem

Business Questions?



BUSINESS Questions :

Can we divide customers to segmentations of risk levels?

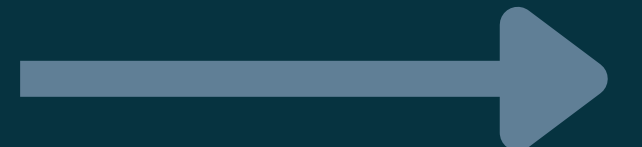
Can we identify customers at risk before they leave?

What factors are associated with customer churn?

Which customer segments have the highest churn rate?

What actions can the company take to retain them?

ACTIONS ?



2- FEATURE SELECTION :

While exploring data I recognized !!

Some attributes that won't be useful for my business analysis

Divided to :

Actionable

VS

Unactionable insights

An insight that the company can take a clear business action on.

EX: Dependent (Make family bundle offers)

An insight that shows a pattern, but does not lead to a clear or practical business action

EX: Education , Martial status

Leading to :

Feature selection

PROJECT OBJECTIVE



Our Business Goal is to :

- ~Identify main factors influencing customer churn.
- ~Finding Business solutions to keep customers with high churn level.



How to do this ?

- ~Compare churn rate across different customer characteristics, such as contract type, tenure, payment method, services, and monthly charges.
- ~Develop targeted retention strategies based on the specific factors driving churn in each segment.

FIRST : EDA (EXPLORATY DATA ANALYSIS) :

churn.info()

Dataset contains :

COLUMNS : 32

ROWS : 1M

TOTAL NULLS : 170193

TOTAL DUPLICATION: 0

IDENTIFIER DUPLICATION: 0

RangeIndex: 1000000 entries, 0 to 999999				
Data columns (total 32 columns):				
#	Column	Non-Null Count		Dtype
----	-----	-----		-----
0	customer_id	1000000	non-null	object
1	signup_date	1000000	non-null	object
2	age	1000000	non-null	int64
3	gender	1000000	non-null	object
4	annual_income	970041	non-null	float64
5	education	1000000	non-null	object
6	marital_status	1000000	non-null	object
7	dependents	1000000	non-null	int64
8	tenure	1000000	non-null	int64
9	contract	1000000	non-null	object
10	payment_method	1000000	non-null	object
11	paperless_billing	1000000	non-null	object
12	senior_citizen	1000000	non-null	int64
13	monthlycharges	1000000	non-null	float64
14	totalcharges	1000000	non-null	float64
15	num_services	1000000	non-null	int64
16	has_phone_service	1000000	non-null	int64
17	has_internet_service	1000000	non-null	int64
18	has_online_security	1000000	non-null	int64
19	has_online_backup	1000000	non-null	int64
20	has_device_protection	1000000	non-null	int64
21	has_tech_support	1000000	non-null	int64
22	has_streaming_tv	1000000	non-null	int64
23	has_streaming_movies	1000000	non-null	int64
24	customer_satisfaction	980079	non-null	float64
25	num_complaints	970094	non-null	float64
26	num_service_calls	1000000	non-null	int64
27	late_payments	1000000	non-null	int64
28	avg_monthly_gb	949988	non-null	float64
29	days_since_last_interaction	1000000	non-null	int64
30	credit_score	959605	non-null	float64
31	churn	1000000	non-null	int64
dtypes: float64(7), int64(17), object(8)				

3- DATA CLEANING :

change dtypes

```
churn['customer_id'] = churn['customer_id'].astype('string')
churn['age'] = churn['age'].astype(int)
churn['contract'] = churn['contract'].astype('string')
```

Handle missing values

```
churn['annual_income'] = churn['annual_income'].fillna(churn['annual_income'].median())
```

Handle inconsistent values

organizing data

```
churn=churn.rename(columns={'tenure': 'tenure(months)'})
```

```
churn['has_internet_service'] = churn['has_internet_service'].str.upper()
churn['has_online_security'] = churn['has_online_security'].str.upper()
churn['has_tech_support'] = churn['has_tech_support'].str.upper()
```

```
[37]: (churn['totalcharges'] < churn['monthlycharges']).sum()
```

```
[37]: 43757
```

4 - FEATURE ENGINEERING :

Some of our added features :

	Risk_Score	Risk_Level
0	0	Low Risk
1	30	Medium Risk
2	0	Low Risk
3	30	Medium Risk
4	0	Low Risk

To divide customers after following patterns to customers who has larger risk of churn

RFM Score	customer segmentation
411	Platinum
123	Bronze
441	Platinum
532	Taitinium
432	Platinum
...	...
452	Platinum
532	Taitinium
143	Bronze
433	Platinum
131	Bronze

To understand customers value to the company

5 - Visualization & Dashborad:



6 - RECCOMENDATION

MEET OUR TEAM

Rahma Alarabi

EDA Presentation &
Data Cleaning

Mariam sheha

Feature Engineering

Mahmoud Zena

Feature Engineering

Lama Elmorshidy

Dashboard & Visualization

Mohameed Abdelrhman

Dashboard & Visualization

Abdelrahman Elhoshy

Recommendation

USED TOOLS :

PYTHON

POWE BI



THANK
YOU

6- Recommendation :

1. The first recommendation is to use a secure and reliable network connection. This is important because the system will be accessed over the internet, and a secure connection is necessary to protect the data and the system from unauthorized access.
2. The second recommendation is to use a secure and reliable web browser. This is important because the system will be accessed through a web browser, and a secure browser is necessary to protect the data and the system from unauthorized access.
3. The third recommendation is to use a secure and reliable operating system. This is important because the system will be installed on an operating system, and a secure OS is necessary to protect the data and the system from unauthorized access.
4. The fourth recommendation is to use a secure and reliable database. This is important because the system will store data in a database, and a secure database is necessary to protect the data and the system from unauthorized access.
5. The fifth recommendation is to use a secure and reliable user interface. This is important because the system will be accessed through a user interface, and a secure UI is necessary to protect the data and the system from unauthorized access.
6. The sixth recommendation is to use a secure and reliable authentication system. This is important because the system will require users to authenticate themselves, and a secure authentication system is necessary to protect the data and the system from unauthorized access.
7. The seventh recommendation is to use a secure and reliable authorization system. This is important because the system will require users to be authorized to access certain data or functions, and a secure authorization system is necessary to protect the data and the system from unauthorized access.
8. The eighth recommendation is to use a secure and reliable logging system. This is important because the system will need to log user activity, and a secure logging system is necessary to protect the data and the system from unauthorized access.
9. The ninth recommendation is to use a secure and reliable backup system. This is important because the system will need to back up data, and a secure backup system is necessary to protect the data and the system from unauthorized access.
10. The tenth recommendation is to use a secure and reliable disaster recovery system. This is important because the system will need to recover from a disaster, and a secure disaster recovery system is necessary to protect the data and the system from unauthorized access.